Shiny App for Differentiability Analysis

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Mathematics in AI

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Problem/Purpose:

For my Project 3 assignment for my Mathematics in AI class, I created an application that showcased when a function would have a derivative at a specified point. This was possible due to the addition of a slider to choose a specified point on a function. The main finding of the application was that it provided a slope on a function's curve that, when interpreted with calculus, proved useful. The equal estimates were still able to grow without any bounds as continuity does not necessarily mean a function is differentiable. My app uses five different function types (quadratic, absolute value, piecewise corner and jump, and cubic root) in combination with tables, graphs, and controls for adjustability to showcase these ideas.

The overall problem with this project objectively is based on a student's understanding of derivatives. An average student would be able to see a function and recognize if it was linear or if it had a smooth curve. The problem comes from when there is an image of a curve or function that does not clearly indicate if and when there are sudden changes in concavity or step changes. The app allows a student to change their viewing scale as well as select specific points of interest to be able to compare their found numerical data with a visual presentation with heightened accuracy.

The five function options reveal three different ways that differentiability can fail. The three ways it fails are discontinuity, unequal slopes, and unbounded slopes. Each of the options will allow a student to have multiple methods of testing the definition of a differentiation and thus, allowing them to compare/contrast the outcomes while keeping the mathematical concepts consistent.

Mathematical Formulation:

We know that a function is differentiable at a specific point when the difference quotient approaches a certain point as increments approaches zero. The definition of a derivative is the slope of the tangent line which is important to know in order to understand that it inherently describes local change. Due to this, we have to evaluate the differentiability of a function at a specified point.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \in \mathbb{R}$$

The limit must agree when x approaches a from the left and from the right. If those approaches produce different slopes, the derivative does not exist. Matching behavior is also insufficient if the slopes increase without bound. These conditions explain why the application examines both sides and why it distinguishes ordinary tangent lines from vertical tangents.

Differentiability implies continuity because the function change equals the increment multiplied by its difference quotient. If the quotient leans toward a finite derivative, that product leans towards zero. Continuity requires the function values to approach the value at the specified point, however it does not require their slopes to approach one finite value. Absolute value at zero puts this idea on display. When a derivative exists, a short portion of the graph can be approximated by a line through the selected point. The approximation becomes more accurate relative to the input change as that change approaches zero. This explains why the tangent overlay is useful only when a finite slope is supported.

Application Numerical Approach:

The app contains a function selector, a displayed interval, a point slider, and a logarithmic step control covering magnitudes from one to one trillionth. A direction control chooses the sign of the secant increment. Optional overlays display the selected secant and, when supported by the diagnostics, an estimated tangent. The interface also reports function values and both sequences of slope estimates. Shiny connects the controls to recalculated outputs through reactive expressions.

The diagnostic rule compares the final left and right estimates using a tolerance of one percent of a scale based on their magnitudes, with a minimum scale of one. It also checks changes among the last three estimates. A positive conclusion requires finite values, agreement, and stability. For the four nonsmooth examples at zero, the code additionally supplies the known mathematical explanation. The resulting label combines numerical evidence with knowledge of the selected function.

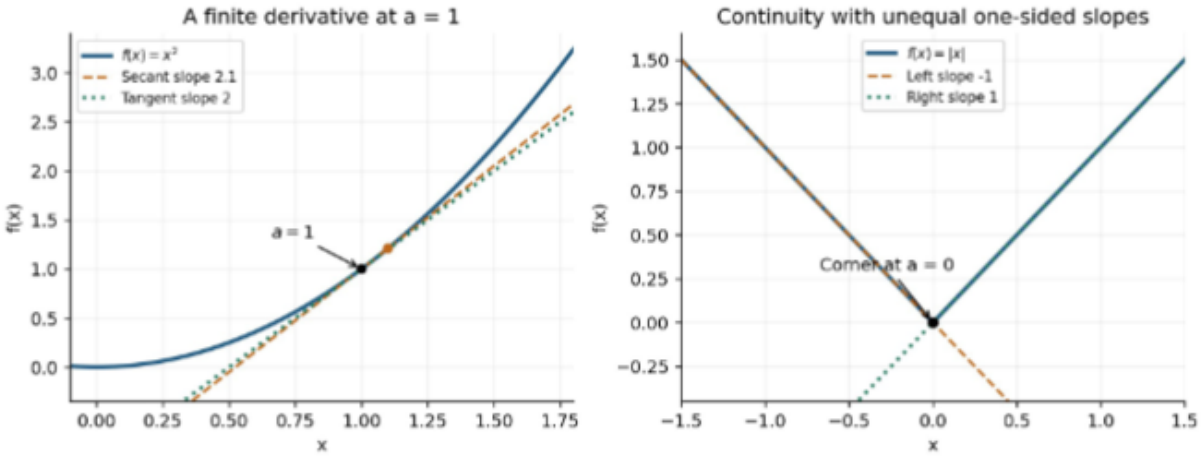
The logarithmic slider makes several orders of magnitude accessible without requiring tiny manual adjustments. Moving it from negative one to negative two changes the initial distance from 0.1 to 0.01. The convergence table then provides finer distances automatically. Comparing rows encourages the user to examine a sequence rather than judge one rounded number.

Interpretation of Results:

The analysis uses the app's default diagnostic settings at the points identified below. The values are generated mathematical data, without external observations or measurements of student performance. The images provided reconstruct the functions and their mathematical features (they are not screenshots of the Shiny interface).

For the quadratic f(x) = x^2, the derivative is 2x everywhere. At a = 1, the exact slope is 2 and the tangent line is y = 2x - 1. Representative left and right estimates are 1.9984375 and 2.0015625. Their proximity to the analytical derivative supports the smooth curve interpretation shown in the given images.

For f(x) = |x| at zero, the graph is continuous because both branches meet at the origin. Its left slope is -1, while its right slope is +1. The disagreement prevents differentiability. Averaging those slopes would produce zero, but that average would not be a derivative. This example shows why the app considers both sides before displaying a tangent estimate.



Vertical Tangents/Piecewise Functions:

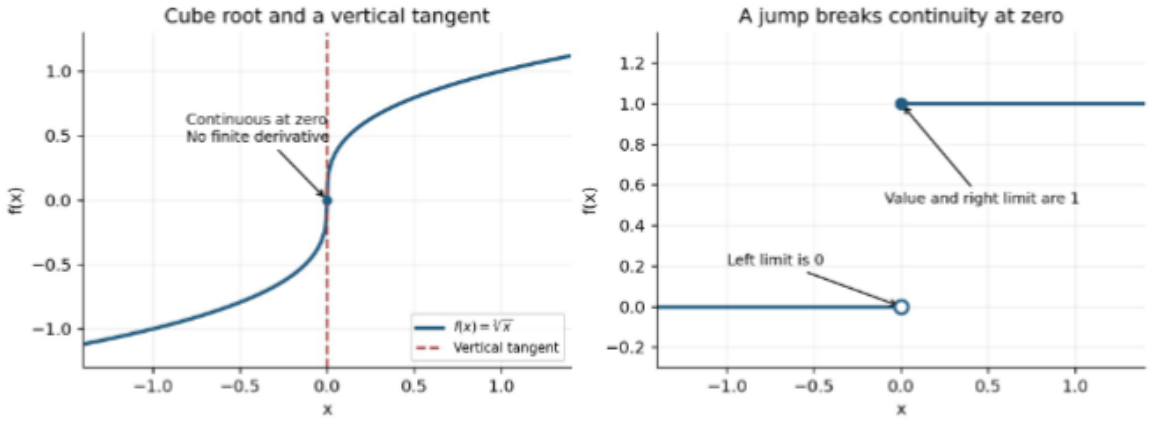
The cube-root function is continuous at zero and has a vertical tangent there. Both sides produce positive slopes that increase without approaching a finite value. A representative pair of estimates is approximately 74.2654 on each side. Their agreement therefore does not establish differentiability. The code evaluates negative inputs using a signed cube-root expression, keeping the graph within the real numbers.

$$\frac{f(x)}{x} = |x|^{-2/3} \to +\infty \quad as\ x \to 0$$

The existence of a geometric tangent does not always imply the existence of an ordinary derivative. A vertical tangent is a line, but its slope cannot be expressed as a finite real number. That is the essential distinction illustrated by the cube-root example.

The piecewise corner equals negative x for x below zero and twice x otherwise. Both branches approach zero, so the function is continuous there. However, the left slope is negative one and the right slope is two. The unequal slopes create a corner. This example reinforces the same derivative criterion with an asymmetric graph.

The jump function equals zero for negative inputs and one otherwise. At zero, its left limit is zero, its right limit is one, and its assigned value is one. The discontinuity rules out differentiability. A right-hand quotient alone can return zero because that branch is constant, demonstrating why evidence from one side can be misleading.



These failures are local. Away from zero, each nonsmooth example has an ordinary derivative on its smooth branches. Absolute value has slope negative one or positive one; the piecewise corner has slope negative one or two. The jump function has derivative zero wherever its graph stays on one constant branch.

Assumptions/Limitations:

Numerical estimates cannot prove a limit exists for every possible function. The app's tolerance can accept small disagreements or leave a mathematically differentiable point unresolved. Its messages should therefore be read alongside the analytical derivative and the graph. Known examples provide useful checks because their exact behavior can be established independently.

Floating-point arithmetic can also affect the calculations. Subtracting nearly equal function values can lose significant digits, and a difference quotient can magnify the resulting rounding error. Agreement between four displayed numbers may reflect their formatting rather than exact equality. The app includes warnings about unreliable calculations, but a warning does not correct the underlying arithmetic.

The graph introduces its own limitations. A viewing interval can conceal a secant endpoint or make a corner difficult to recognize. The examples are defined for every real input, so calculations outside the visible window remain mathematically meaningful. An extension to functions with restricted domains would need additional checks before evaluating nearby inputs.

This study evaluates mathematical consistency and the clarity of the explanation. It does not establish that the app improves learning outcomes. Demonstrating that benefit would require a separate study with suitable assessment methods and participants.

The function menu covers a deliberately limited collection of examples. Oscillating functions can also lack derivatives, even when their graphs have no single corner or jump. A general diagnostic tool would therefore need broader examples and stronger analytical checks before its conclusions could be applied universally.